

Subject Description Form

Subject Code	ITC6982
Subject Title	Biomimicry in Fashion
Credit Value	3
Level	3
Pre-requisite / Co-requisite / Exclusion	N/A
Objectives	Guide students to carry out interdisciplinary and cost-effective researches, and to resolve problems with adaptable and sustainable approaches.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - understand fashion from a view of adaptable and sustainable system; - implement an interdisciplinary research based on a holistic understanding of nature; - consciously and actively align their researches to a evolved nature way; - creatively design sophisticated green approaches to solve human problems; - continuously develop their knowledge and ability of harmonizing artificial systems with living nature.
Subject Synopsis/ Indicative Syllabus)	- Principle of biomimicry: adaptable & sustainable approach. - Fashion with diverse fibers in nature. - Fashion with functional surfaces in nature. - Fashion with natural thermal insulation. - Fashion with natural optical systems. - Complexity in fashion biomimicry: artificial vs. living.
Teaching/Learning Methodology	Combination of group discussion and self-reading, reference investigation and hand practice, case study and presentation, inspiration and reflection.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			a	b	c	d	e
	1. Continuous assess.	40	✓	✓	✓	✓	✓
	2. Open-book exam	60	✓	✓	✓	✓	✓
	Total	100%					
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Through group assignments & presentations, and individual final exam, all the above learning outcomes can be assessed based on students' behaviors.							
Student Study Effort Expected	Class contact:						
	▪ Group discussion						24 Hrs.
	▪ Tutorial						12 Hrs.
	Other student study effort:						
	▪ assignments & presentations						44 Hrs.
	▪ reference investigation						40 Hrs.
	Total student study effort						120 Hrs.
Reading List and References	Bhushan, B. 2009 Biomimetics: lessons from nature—an overview. Phil. Trans. R. Soc. A 367, 1445–1486. Dawson, C., Vincent, J. F. V. & Rocca, A. 1997 How pine cones open. Nature 390, 668. Koch, K., Bhushan, B. & Barthlott, W. 2009 Multifunctional surface structures of plants: an inspiration for biomimetics. Prog. Mater. Sci. 54, 137–178. Vukusic, P. & Sambles, J. R. 2003 Photonic structures in biology. Nature 424, 852–855.						